

• **Teams may have the following:**

- One stand-alone, non-programmable, non-graphing calculator.
- One hard copy 8.5" x 11" sheets of paper with information from any source. **Electronic notes are fine to use.**

- **Partial credit will be awarded accordingly.**
- **(Free Responses) Make sure that you are answering a question on the corresponding space given on the answer sheet!**
- **If you are not certain as to what you should be doing, or if a question does not make sense to you, ask the event supervisor what to do.**
- **For fill-in-the-blank questions, assume your answer to be singular unless you're given different instructions.**
- **Be realistic in all of your answers. Think biologically and environmentally!**
- **You have 50 minutes to complete this test. Good luck!**

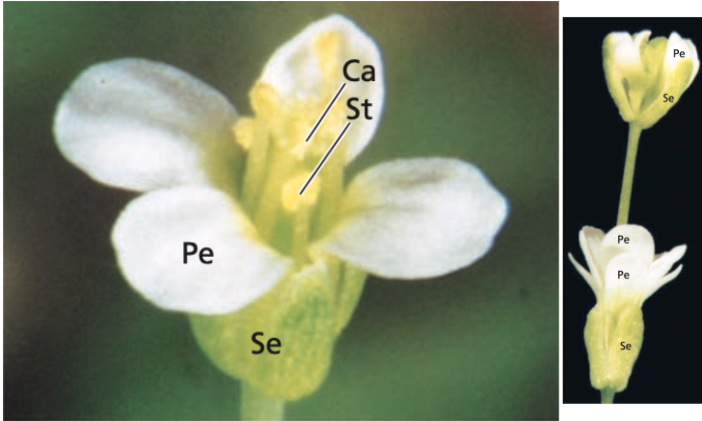
1. (1.00 pts)

In Theophrastus' famous book, *Enquiry Into Plants*, one of the 10 books on the natural history of plants describes it's medicinal uses. For example, hemlock is a poison that gives a painless death; pepper is its antidote. Which of the following plants can be used to expel a woman's placenta?

- ☐ A) Strychnos
- ☐ B) Peony
- ☐ C) Frankincense
- ☒ D) Birthwort

2. (1.00 pts)

Look at the images provided above (left, normal; right, abnormal). According to the ABC Hypothesis in flower development, this abnormal *Arabidopsis* flower seems to have an internal flower replacing carpels in normal plants, meaning that there is a mutation in which of the following genes.



- ☐ A) A
- ☐ B) B
- ☒ C) C
- ☐ D) None

3. (1.00 pts)

Which of the following phytopathogenic diseases is on a tree in the picture below?



- ☐ A) Apiosporina morbosia
- ☒ B) Thousand Cankers
- ☐ C) Black Knot
- ☐ D) Anthracnose

4. (1.00 pts) Cedar-apple rust is an example of a disease caused by a fungus with a polycyclic life cycle.

- ☐ True
- ☒ False

5. (1.00 pts) What is the production of fruit without any seeds called?

Parthenocarpy

6. (1.00 pts) A. tumefaciens genes force plant cells to produce which of the following plant hormones?

- ☒ A) Auxin and Cytokinin
- ☐ B) Auxin and Gibberellin
- ☐ C) Cytokinin and Ethylene
- ☐ D) Gibberellin and Testosterone
- ☐ E) Absciscic acid and Ethylene

7. (1.00 pts) Powdery mildew can be wiped off of the surface of a plant, making it easy to control without the use of a fungicide.

- ☒ True
- ☐ False

8. (1.00 pts) Which of the following is **not** an agent that interferes with oxidation phosphorylation or photophosphorylation?

- ☐ A) Myxothiazol
- ☐ B) Amytal
- ☐ C) Rotenone
- ☒ D) Ubiquinone

9. (1.00 pts) What part of a plant's chloroplast does the Calvin Cycle take place?

- ☐ A) Inner membrane

- ☐ B) Outer membrane
- ☐ C) Thylakoid
- ☒ D) Stroma

10. (1.00 pts)

In [answer], CO₂ is first fixed in mesophyll cells into oxaloacetate then malate, which then passes into bundle-sheath cells and releases CO₂ in high concentrations.

- ☐ A) C₂ plants
- ☐ B) C₃ plants
- ☒ C) C₄ plants
- ☐ D) CAM plants

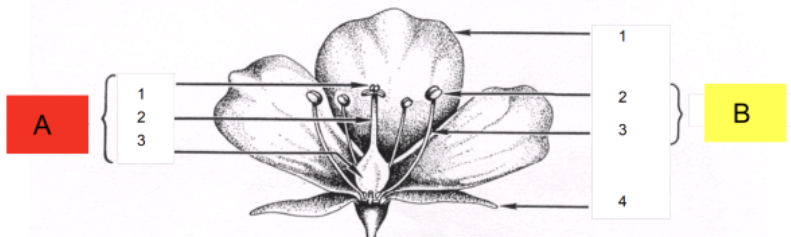
11. (1.00 pts) Where would one look for oxygen production in photosynthesis?

- ☐ A) Hack and Slack cycle
- ☐ B) Cyclic phosphorylation
- ☒ C) Noncyclic phosphorylation
- ☐ D) Calvin Cycle

12. (1.00 pts) Which of the following enzymes helps an excited Photosystem I (PSI) pass its electrons from ferredoxin to NADP⁺.

- ☐ A) PQa
- ☒ B) FNR
- ☐ C) PQb
- ☐ D) Plastocyanin

13. (1.00 pts) The following question will test your knowledge of this flower's anatomical parts. Read all of the provided choices then select the **incorrect** answer.



- ☐ A) A3; contains one or more ovules that develop into seeds once fertilized.
- ☐ B) B2; male reproductive part which contains pollen.
- ☐ C) B4; collectively called the calyx.
- ☐ D) A1; a source of nectar that attracts insects and birds.
- ☒ E) A2; often sticky from which pollen germinates.

14. (1.00 pts) Inflorescence is a botanical term used to describe the arrangement of different plants, in this case, flowers. What type of inflorescence flower is this?



- ☐ A) Panicle
- ☐ B) Raceme
- ☐ C) Compound cyme
- ☒ D) Compound umbel
- ☐ E) Corymb

15. (1.00 pts) Compare the plant in the previous question to a cycad. Which of the following scientific classifications is unique to a cycad?

- ☐ A) Eukaryota
- ☐ B) Spermatophyte
- ☒ C) Gymnospermae
- ☐ D) Trachophytes
- ☐ E) Eudicots

16. (1.00 pts) Plastic mulches are not effective for evaporation control because they are not biodegradable.

- ☐ True
- ☒ False

17. (1.00 pts) Which of the following viruses is used for the bio-control of insect pests in plants?

- ☐ A) Rice Tungro Virus
- ☒ B) Nuclear Polyhedrosis Virus
- ☐ C) Cucumber Mosaic Virus
- ☐ D) Tomato Mosaic Virus

18. (1.00 pts) Which of the following is not an adaptation of a hydrophyte?

- ☐ A) There exists a poorly developed root system.
- ☒ B) There is less parenchyma in the stem.
- ☐ C) Stomata found on the upper leaf surface.
- ☐ D) There are fewer cuticles over the epidermis by comparison.

19. (1.00 pts) Which of the following is not an adaptation of xerophytic plants?

- ☐ A) Presence of palisade-like tissue in the stem
- ☐ B) Presence of thick cuticles
- ☐ C) Adopts the CAM cycle for photosynthesis

- ☒ D) Large leaves are present

20. (1.00 pts)

Black mustard seeds (*Brassica juncea*) are edible and used as the phytoremediation agent for the removal of which of the following heavy metals found in the soil?

- ☒ A) Lead
☐ B) Cadmium
☐ C) Nickel
☐ D) Arsenic
☐ E) Copper

21. (1.00 pts) Which of the following routes allow water and minerals to diffuse into the cortex along the matrix of walls and extracellular spaces?

- ☐ A) Symplastic
☐ B) Transmembrane
☒ C) Apoplastic
☐ D) Xylemic
☐ E) Vascular

22. (1.00 pts) This discolored leaf exhibits a disease that is common in acidic or sandy soils, indicating which macronutrient deficiency?



- ☐ A) Mg^{2+}
☐ B) K^{+}
☐ C) Ca^{2+}
☒ D) SO_4^{2-}

23. (1.00 pts) The Mistletoe plant would be most closely associated with which of the following symbiotic relationships?

- ☐ A) Commensalism
☐ B) Mutualism
☒ C) Parasitism
☐ D) Recidivism
☐ E) Amensalism

24. (1.00 pts) Plants that grow under direct sunlight and called [Answer 1], and those growing under shade are called [Answer 2].

- ☐ A) Heliophytes and Psamophytes
☐ B) Psamophytes and Heliophytes
☐ C) Sciophytes and Psamophytes
☒ D) Heliophytes and Sciophytes

25. (1.00 pts)

In callus culture, higher concentrations of auxin rather than cytokinin induce the formation of which plant structure?

- ☐ A) Adventitious Shoot
- ☒ B) Adventitious Root
- ☐ C) Both Root and Shoot
- ☐ D) None of the Above

26. (1.00 pts)

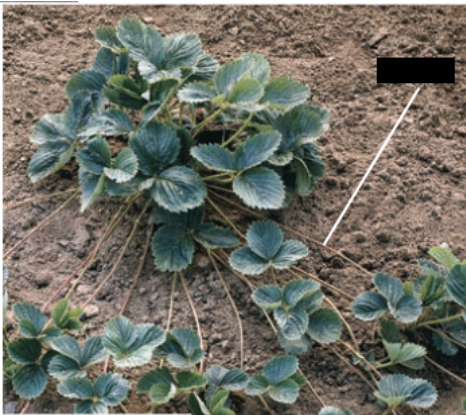
Mangrove trees inhabiting low tides and growing in oxygen-poor waters develop unique root systems such that their roots appear in the air. What are these roots known as?

- ☐ A) Buttress roots
- ☐ B) Strangling aerial roots
- ☒ C) Pneumatophores
- ☐ D) Prop roots

27. (2.00 pts)

The evolutionary adaptation of stems by plants is quite unique. Whether for food storage or asexual reproduction, plants develop modified stems that are often mistaken for roots.

- Identify the following modified stem in this image below:



(Mark **ALL** correct answers)

- ☒ A) Stolons
- ☐ B) Tendrils
- ☒ C) Runners
- ☐ D) Suckers
- ☐ E) Offsets
- ☐ F) Bulbs

28. (1.00 pts) Lateral plant roots stem from which of the following plant structures?

- ☐ A) Pith
- ☐ B) Endodermis
- ☐ C) Cortex
- ☒ D) Pericycle

29. (1.00 pts) Narrowed xylem cells with lignified walls that help with water transportation.

- ☐ A) Endodermis
- ☐ B) Collenchyma
- ☐ C) Sclerenchyma
- ☒ D) Tracheids

- ☐ E) Periderm

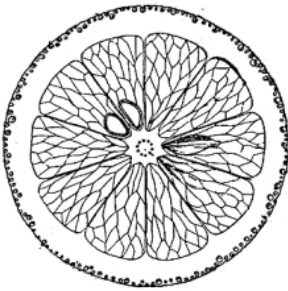
30. (2.00 pts) Which of the following describes connation in flowers?

- Select the correct answer(s):
 1. Styles connected to stamen
 2. Petals connected to each other
 3. Pistil connected to stamen
 4. Carpels connected to each other

(Mark **ALL** correct answers)

- ☐ A) 1
- ☐ B) 2,3
- ☐ C) 1,4
- ☒ D) 2,4
- ☐ E) 4

31. (1.00 pts) Which of the following correctly identifies the taxonomical category of the following image?



- ☐ A) Achenes
- ☐ B) Drupe
- ☐ C) Pepo
- ☒ D) Hesperidium
- ☐ E) Cirkusis
- ☐ F) Pome

32. (1.00 pts) Which of the following came second in plant evolution?

- ☐ A) Non-vascular plants
- ☐ B) Conifers
- ☒ C) Seedless vascular plants
- ☐ D) Flowering plants

33. (2.00 pts) Plants that can switch between C3 and CAM mode are called _____. **Capitalize the first word. Capitalize the entire second word (acronym).**

Facultative

CAM

34. (1.00 pts)

A microbe found on a plant with *Erwinia carotovora* may have caused a certain disease but you cannot find published information verifying this. Which procedure do you follow to determine whether this pathogen caused that disease?

- ☐ A) Fungal determinant.
- ☐ B) Karnal's rules.
- ☐ C) Draw numbers out of a hat.
- ☒ D) Koch's postulates.

- ☐ E) Moist chamber.

35. (2.00 pts) Which of the following diseases does the previous question's microbe cause?

(Mark **ALL** correct answers)

- ☐ A) Aster yellows
- ☒ B) Soft rot
- ☐ C) Apple scab
- ☒ D) Slime flux
- ☐ E) Corn smut

36. (1.00 pts) Fireblight initially evolved in which continent?

- ☐ A) Asia
- ☒ B) North America
- ☐ C) South America
- ☐ D) Europe

37. (1.00 pts) Biodiesel is obtainable from which plant?

- ☐ A) Prosopis
- ☐ B) Catharanthus
- ☐ C) Calotropis
- ☒ D) Jatropha curcas

38. (1.00 pts) A biodiversity hotspot is characterized by which of the following concepts?

- ☐ A) Endemic plants
- ☐ B) Threat perception
- ☒ C) Endemic flowering plants
- ☐ D) Species flowering plants

39. (1.00 pts) Which of the following ecosystem has the longest energy transfer time?

- ☐ A) Open ocean
- ☐ B) Tropical rain
- ☒ C) Temperate deciduous
- ☐ D) Desert

40. (1.00 pts)

Gibberellic Acid (GA) controls seed germination by directing breakdown of the stored starch. Which of the following barley seed tissues does the α -amylase gene is induced in respond to GA?

- ☐ A) Endosperm
- ☐ B) Coleoptile
- ☒ C) Aleurone layer
- ☐ D) Embryo

41. (1.00 pts)

Nitrogen fixation is a process where nitrogen gas is converted into ammonia. One of the key enzymes in this process is nitrogenase. Which of the following statements best highlights the production and activity of nitrogenase?

1. Nitrogen fixation through nitrogenase is an energetically expensive process.
2. Nitrogenase encoding gene is under a constitutive promoter.
3. Nitrogenase is highly sensitive to oxygen.
4. Endogenous availability of the cofactor of nitrogenase enzyme is very low.

- ☐ A) 1 and 2
- ☒ B) 1 and 3
- ☐ C) 2 and 3
- ☐ D) 2 and 4

42. (1.00 pts) The bending of an oat coleoptile tip towards the source of unilateral light of wavelength 454 nm is due to which of the following?

- ☐ A) Synthesis of auxin in the shaded area.
- ☒ B) Lateral distribution of auxin toward shaded area
- ☐ C) Degradation of auxin in the light area.
- ☐ D) Polar transport of auxin.

43. (1.00 pts) The hydrostatic opening of stomata is due to which of the following?

- ☐ A) Movement of K⁺ ions into guard cells.
- ☒ B) Low water potential in mesophyll cells.
- ☐ C) Movement of water into guard cells.
- ☐ D) Decreased concentration of ABA.

44. (1.00 pts) Which of the following are regulators of plant circadian rhythms?

- ☐ A) Phototropins
- ☐ B) Cryptophores
- ☐ C) Phycobilins
- ☒ D) Phytochromes

45. (1.00 pts) What type of fruit is a pineapple?

- ☐ A) Berry
- ☐ B) Aggregate
- ☐ C) Dehiscent
- ☒ D) Multiple

46. (1.00 pts) What are the paired outgrowths typically found alongside the petiole attachment to the stem called?

- ☒ A) Stipules
- ☐ B) Crazy twig
- ☐ C) Areoles
- ☐ D) Lenticels

47. (1.00 pts) Certain plants have an anatomical feature called a holdfast. Such a plant would be associated with which biome?

- ☐ A) Desert
- ☒ B) Ocean
- ☐ C) Tropical rainforest
- ☐ D) Tundra

48. (1.00 pts) Abscission in a plant would be associated with which of the following process?

- ☐ A) Germination
- ☐ B) Reproduction
- ☐ C) Growth
- ☒ D) Defoliation

49. (1.00 pts) Which of the following is an example of an organ of perennation in plants?

- ☒ A) Rhizome
- ☐ B) Seed
- ☐ C) Petal of a flower
- ☐ D) Calyx of a flower

50. (1.00 pts) Which of the following statements is correct?

- ☐ A) Plants absorb carbon dioxide and only release oxygen.
- ☐ B) Plants absorb carbon dioxide, and release oxygen and carbon dioxide.
- ☒ C) Plants absorb carbon dioxide and oxygen, and release oxygen and carbon dioxide.
- ☐ D) Plants absorb oxygen and only release carbon dioxide.

51. (1.00 pts) The embryonic portion of the seed is called the radicle is associated with which of the following plant parts?

- ☐ A) Fruiting body
- ☒ B) Root
- ☐ C) Leaf
- ☐ D) Stem

52. (1.00 pts) What plant is known as the smallest flowering plant? **Hint:** Has two tiny vascular roots. Capitalize the first letter of the first word. One-word answer.

Duckweed

53. (1.00 pts) Thorns are found on many different plants but biologically are not modified leaves.

- ☐ True
- ☒ False

54. (1.00 pts) What is the symbiotic relationship between various fungi and plant roots called? Capitalize the first letter of the first word. One-word answer.

Mycorrhizae

55. (1.00 pts) The relationship described in the previous question occurs in what percentage of plants?

- ☐ A) 50%
- ☐ B) 70%
- ☒ C) 80%
- ☐ D) 90%

56. (1.00 pts) Some plants produce an underground stem that is almost entirely stem tissue. Which of the following option refers to such a growth?

- ☒ A) Corm
- ☐ B) Rhizome
- ☐ C) Stolon
- ☐ D) Tuber

57. (1.00 pts) Which of the following was not restricted to the Western Hemisphere until 1942?

- ☒ A) Banana
- ☐ B) Cacao
- ☐ C) Potato
- ☐ D) Pepper

58. (1.00 pts)

The photosynthetic process can only make use of a fraction of the total energy in sunlight. For a "typical" plant, how much of sunlight is in the photosynthetically active wavelength range?

- ☐ A) 65%
- ☐ B) 55%
- ☒ C) 45%
- ☐ D) 35%

59. (1.00 pts) Which macroscopic group of marine plants is the most diverse and abundant?

- ☐ A) Chrysophyta
- ☐ B) Phaeophyta
- ☐ C) Pyrrophyta
- ☒ D) Rhodophyta

60. (1.00 pts) Sedges superficially resemble grasses. Which of the following statements does not accurately describe the differences between grasses and sedges?

- ☐ A) Sedges have triangular stems with leaves that are arranged in three ranks.
- ☒ B) Sedges have hollow stems.
- ☐ C) Sedge leaves confined to the base and/or top of the plant.
- ☐ D) Sedges have specialized flowering structures.

61. (1.00 pts) Periphyton is a type/group of plants that would be associated with which of the following settings?

- ☐ A) Sand dunes
- ☐ B) Tropical/Semitropical forests
- ☒ C) Rivers/Streams
- ☐ D) Tundras

62. (1.00 pts)

Many trees, mostly gymnosperms, do not release their seeds until exposed to high heat, such as a fire. Such a cone is described as? **Capitalize the first letter of the first word. One-word answer (adjective).**

Serotinous

63. (1.00 pts) Which color light is the most important for plant growth and flowering?

- ☒ A) Blue and Red
- ☐ B) Green and Blue
- ☐ C) Red and Green
- ☐ D) Orange and Red

64. (1.00 pts) Which of the following reasons correctly describes why coffee plants produce caffeine?

- ☐ A) As an energy storage mechanism.
- ☒ B) As a pesticide to ward off insects.
- ☐ C) As a means to help survive extended dry periods.
- ☐ D) As an astringent to discourage grazing animals.

65. (1.00 pts)

If you were a chiropterophilus plant, you would have adapted yourself to attract the attention of what organism for pollination? Capitalize the first letter of the first word. One-word answer.

Bat

66. (1.00 pts) Any response of a plant to the relative lengths of daylight and darkness. Capitalize the first letter of the first word. One-word answer.

Photoperiodism

67. (1.00 pts) Which of the following trees are well known for allelopathy?

- ☐ A) Oak
- ☒ B) Walnut
- ☐ C) Beech
- ☐ D) Pecan

68. (1.00 pts)

California is one of the nation's largest agricultural producers. Agriculture in California is faced with the constant (and growing) challenge of limited water supply. Which of these crops is the largest consumer of water (total aggregate volume) in that state?

- ☐ A) Rice
- ☐ B) Vineyards/Wine
- ☐ C) Citrus
- ☒ D) Almonds/Pistachios

69. (1.00 pts) What are the oldest living trees on Earth called?

- ☒ A) Bristlecone Pines
- ☐ B) Gingko
- ☐ C) California Sycamore
- ☐ D) Utah Juniper

70. (1.00 pts) Which soil is rich in organic debris?

- ☐ A) Loam soil
- ☒ B) Humus soil
- ☐ C) Saprophyte
- ☐ D) Sandy soil

71. (1.00 pts) Which of the following is not considered to be a true fruit?

- ☐ A) Date
- ☐ B) Plum
- ☒ C) Apple
- ☐ D) Grape

72. (1.00 pts) What is the term used to describe "land plants"? Capitalize the first letter of the first word. One-word plural answer.

Embryophytes

73. (1.00 pts)

What group of plants is distinguishable from all other plant groups by the complete absence of stomata? **Capitalize the first letter of the first word. One-word plural answer.**

Liverworts

74. (1.00 pts) Which one of the following is not associated with stomata?

- ☐ A) Water
- ☐ B) Oxygen
- ☐ C) Carbon Dioxide
- ☒ D) Hydrogen

75. (1.00 pts) Which of the following are not a basic growth form of lichen?

- ☐ A) Foliose
- ☒ B) Fructose
- ☐ C) Crustose
- ☐ D) Glucose
- ☐ E) Fruticose

76. (1.00 pts) What is the name for a fire that runs through the tops of living trees or bushes? **Capitalize the first letter of the first word. One-word answer.**

Crown

77. (1.00 pts) What type of "fruit" structure is produced by an American Elm? **Capitalize the first letter of the first word. One-word answer.**

Samara

78. (1.00 pts)

Dead empty tubes that make up the largest component of conifer wood (90-94% of volume) are known as what? **Capitalize the first letter of the first word. One-word plural answer.**

Tracheids

79. (1.00 pts) How many molecules of carbon dioxide are needed in photosynthesis to produce one glucose molecule?

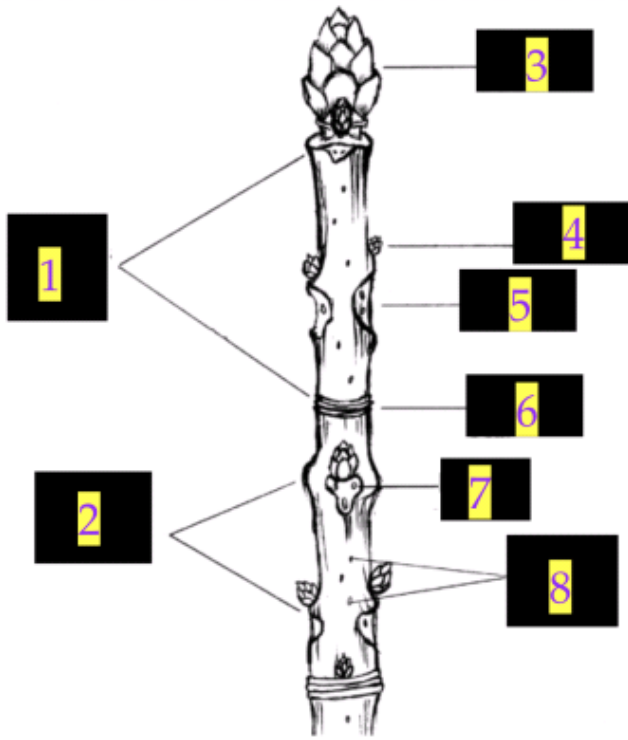
- ☐ A) 4
- ☒ B) 6
- ☐ C) 2
- ☐ D) 12
- ☐ E) 3

80. (8.00 pts)

Stem Characteristics

Plant Anatomy

- From the left column down then right column down, identify the following stem parts which are numerically classified. **Number each** answer in order to make grading more convenient.



Hint: Answer for Number 1 = "Last year's growth"

Expected Answer: +1 per correct answer and in the following sequence: Last Year's Growth, Internode, Terminal Bud, Axillary Bud, Leaf Scar, Bud Scale Scar (or Bud Scar), Node, Lenticels.

BeLeaf In Yourselfs

Plant Anatomy

The following questions will test your knowledge on leaves (e.g. shapes, margins, tips, bases). Answer them to the best of your ability. **Capitalize the first letter of the first word for each answer.** For example, **Bipinnately compound** rather than **Bipinnately Compound**.

81. (1.00 pts) Which of the following leaf bases describes the leaf image below?



- ☐ A) Peltate
- ☐ B) Truncate
- ☐ C) Dentate
- ☒ D) Sagittate
- ☐ E) Hastate

82. (1.00 pts) Type the answer that describes the leaf margins in this picture below. Remember to capitalize the first letter only. One-word answer.



Serrate

83. (2.00 pts) Identify this leaf shape followed by its venation. First answer should be the leaf shape. Second answer should be its venation type.

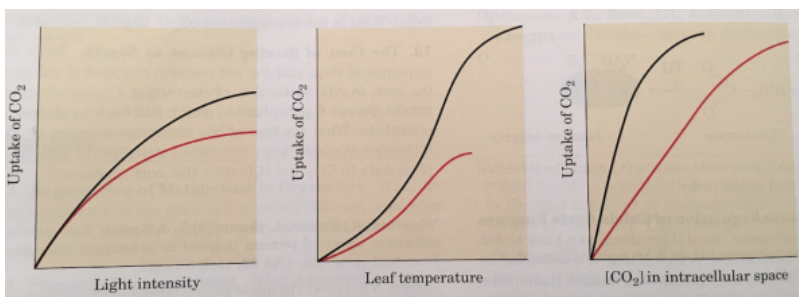


Ovate

Pinnate

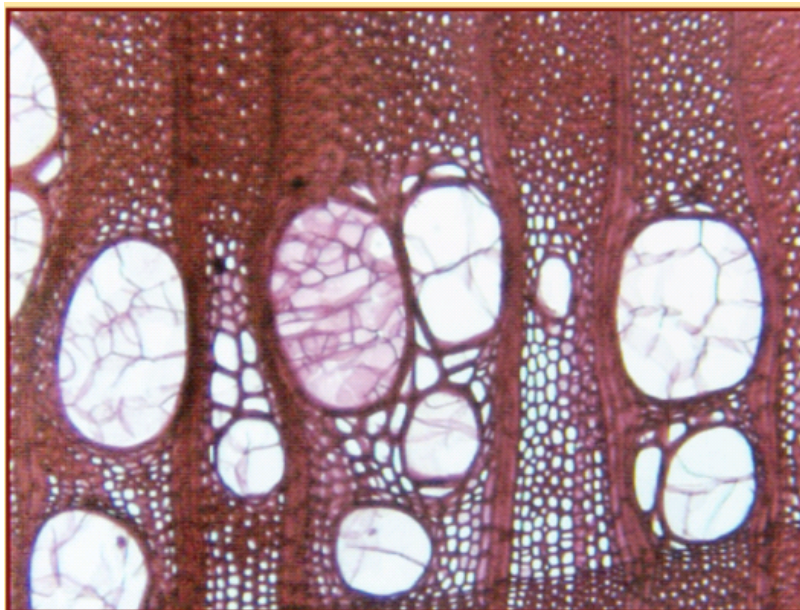
84. (7.00 pts) The plant genus *Alternanthera* includes some C3 and C4 species.

- From the three graphs below, **identify** and **justify** which species is a C3 plant and which is a C4 plant. Species 1, upper curve; Species 2, lower curve.
- Name one C3, C4, and CAM plant.



Expected Answer: +1 for correct identification of C3 plant (Species 2). +1 for correct identification of C4 plant (Species 1). +2 for valid justification. For example, photorespiration (loss to process fixate carbon for subsequent carbohydrate synthesis): C3 plants are less efficient than C4 because as CO₂ levels increase, photorespiration is suppressed. +1 for

85. (10.00 pts) Below is a microscopic transverse image of the native black locust (*Robinia pseudoacacia*) displaying a plant condition.



- **Name** the growths around these vessels and **identify** their relative location.
- **Describe** how these growths are formed.
- **Reason** why these growths are formed.
- **What** role do ergastic substances play? **Include** two ergastic substances in your answer.
- **Describe** how this relates to CODIT.

Expected Answer: +1 for identifying the tyloses growths. +1 for identifying the relative location. +2 for valid description of tyloses formation. +2 for valid reason; environmental response. +1 for valid reason of ergastic substance use. +1 for identifying an ergastic substance. Maximum of 2 obtainable points. +1 for correctly connecting this to CODIT.

86. (2.00 pts) Below is an image of the effects of **Dutch Elm Disease** spread by elm beetles.



- In your own words, **describe** two preventable management treatments that can control the spread of **Dutch Elm Disease**.

Expected Answer: +0.5 if competitors just identify a valid treatment. Maximum of 1 obtainable point. OR +1 for describing a valid treatment. Maximum of 2 obtainable points. Again, refer to the more descriptive Answer Key (Google Docs). Mechanical, Chemical, Biological.

87. (4.00 pts)

Seed dormancy is caused by inner or outer embryonic conditions activated by survival mechanisms (e.g. physiological, mechanical, chemical). Absciscic acid is a chemical inhibitor responsible for delayed seed germination. Well, actually, the ratio of absciscic acid to gibberellins determines whether a seed germinates or remains dormant.

- Describe two human practices of breaking seed dormancy to promote and ensure successful germination?

Expected Answer: +1 if competitors simply identify a human practice. Maximum of 2 obtainable points. +2 for describing valid practice. Maximum of 4 obtainable points. For example, in stratification, gardeners nudge the seeds in the right direction stratification by putting seeds in a refrigerator for a month or in a heated propagator. In scarification, gardeners are more aggressive. They rub the seed coat with a file or with small seeds placed and shaken in a screw-lid jar lined with sandpaper. Cutting, chipping, or pin-pricking the seed coat are other options. In soaking, this can remove natural chemical inhibitors like absciscic acid.

88. (12.00 pts)

C4 and CAM Plants

Plant Photosynthetic Pathways

Evolution helps us explain how succulent cacti and pineapple continue to thrive on counterproductive (arid and barren) environments.

On a hot, dry day, most plants close their stomata, a response that conserves water but also reduces CO₂ levels. With stomata even partially closed, CO₂ concentrations begin to decrease in the air spaces within the leaf, and the concentration of O₂ released from the light reactions begins to increase. These conditions within the leaf favor an apparently wasteful process called photorespiration.

- **Explain** how the C4 and CAM pathways are two evolutionary solutions of maintaining photosynthesis with stomata partially or completely closed on hot, dry days? Your response must have the following terms: **CO₂, mesophyll cells, bundle-sheath cells, Calvin Cycle, PEP Carboxylase**. Hint: Identify a C4 and CAM plant then use them in your description.
- **Identify** two biomes that C4 plants dominate and **describe** why?

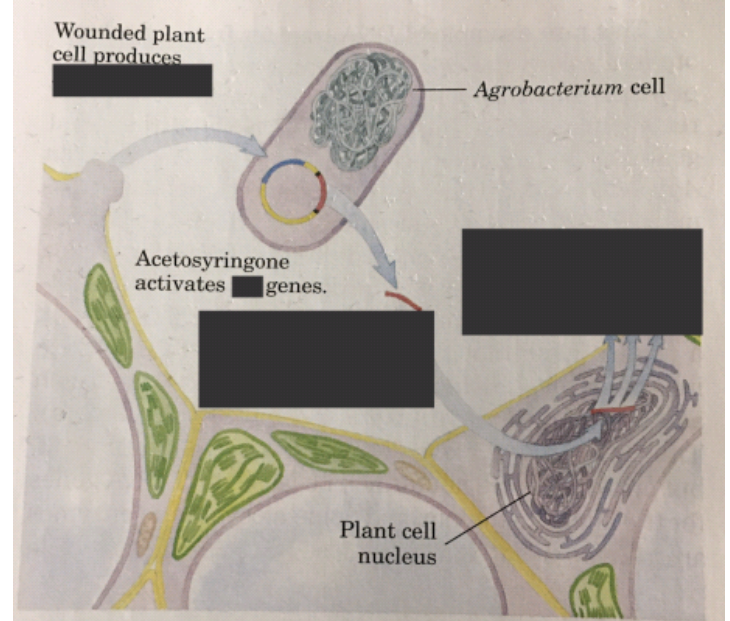
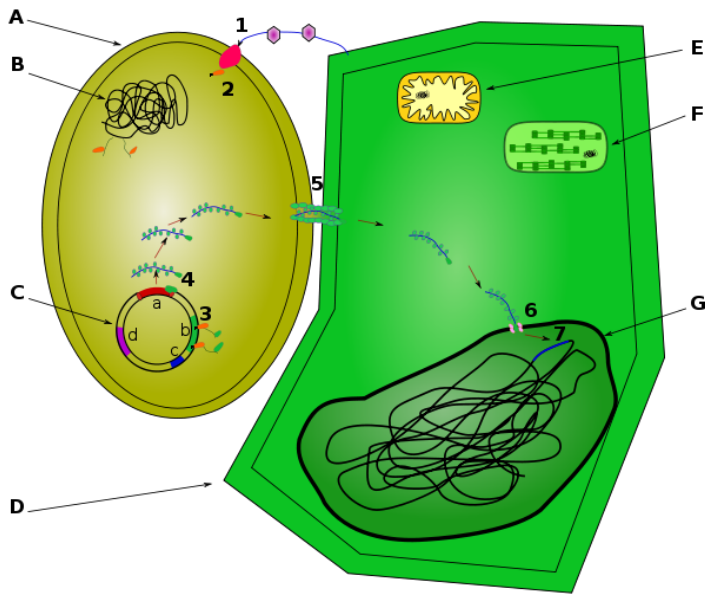
Expected Answer: View answer key.

A Bacterial Plant Parasite Aids Cloning In Plants

Plant Genetics and Disease

Researchers are looking into an important and adaptable ally, *Agrobacterium tumefaciens*, because of its ability to invade plants at wounded sites, then transform nearby plant cell chromosomes to synthesize plant hormones for bacterial growth and nutrition. Research can unlock enormous agricultural implications that give alternative possibilities to nutrition and resistance to environmental stresses (e.g. insects, diseases, cold, drought). The following questions will test your knowledge of plant genetics and diseases.

- [Here are some supplementary images!](#)



89. (17.00 pts) Agrobacterium Engineering Is The Future

- Carefully read the following statements regarding *Agrobacterium* in plant transformation to answer the first short answer question:

- Region A in Ti Plasmid is responsible for replication.
- Region D in Ti Plasmid is responsible for virulence.
- A. tumefaciens* causes hairy root disease and *A. rhizogenes* cause crown gall disease.
- Oncogenic (onc) region in T-DNA is responsible for unusual amino acid synthesis.

- Identify the correct statement(s) by giving **numeric answer(s)**. For example, your answer may be 3, or, 1 and 2, or, 2 and 4. **No partial credit!**
- Identify what phenolic compound is released by a wounded plant cell.
- Describe the DNA transfer to plant cells by bacterial parasites which are the steps in the image provided. Hint: **There are five steps.**
- More specifically, what are the five main steps that vir proteins take to transfer copies of Agro T-DNA into a plant's chromosome? Describe two.
- Identify three types of metabolites converted for the bacterium from plant metabolites through enzymes?

Expected Answer: +1 for identifying the correct answer (Option 2 only) Would be option 3 too but *A. tumefaciens* causes crown gall disease in plants, not hairy root disease. *A. rhizogenes* causes the latter disease. +1 for identifying the correct phenolic compound (Acetosyringone). +1 for valid description of one step. Maximum of 5 obtainable points. +1 for identifying correct step. Maximum of 5 obtainable points. +1 for correct description of a correct step. Maximum of 2 obtainable points. +1 for identifying correct metabolite. Maximum of 3 obtainable points. Auxin, Cytokines, and Opines.

90. (9.00 pts) Plant Evolution

Figure 1 -> Everyday Plants

The following question will test your knowledge on how plants evolved from the following image, Figure 1:



- What is Figure 1?
- Describe two of the unique characteristics that are shared by Figure 1 and plants, which are not shared by other photosynthetic protists.
- Describe two advantages of photosynthetic autotrophs to being in shallow waters and potentially moving onto land?
- What were two evolutionary challenges that Figure 1 and their early descendants had to overcome?
- What are sporopollenin polymers? How did it help the ancestors of Figure 1 spend time on land?

Expected Answer:

91. (12.00 pts)

Plant Diversity

Plant Reproduction

The life cycles of all plants alternate between two generations of distinct multicellular organisms: gametophytes and sporophytes. As shown in the diagrams below, each generation gives rise to the other, a process that is called **alternation of generations**. This type of reproductive cycle evolved in various groups of algae but does not occur in the algae most closely related to plants. Take care not to confuse the alternation of generations in plants with the haploid and diploid stages in the life cycles of other sexually reproducing organisms. The following questions will test your knowledge on non-vascular and vascular plant reproductive systems.

Below are the two diagrams, or plant life cycles:

Figure 1:

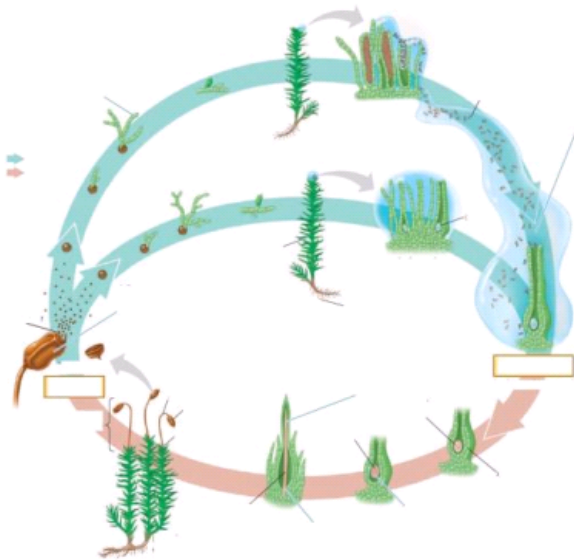


Figure 2:

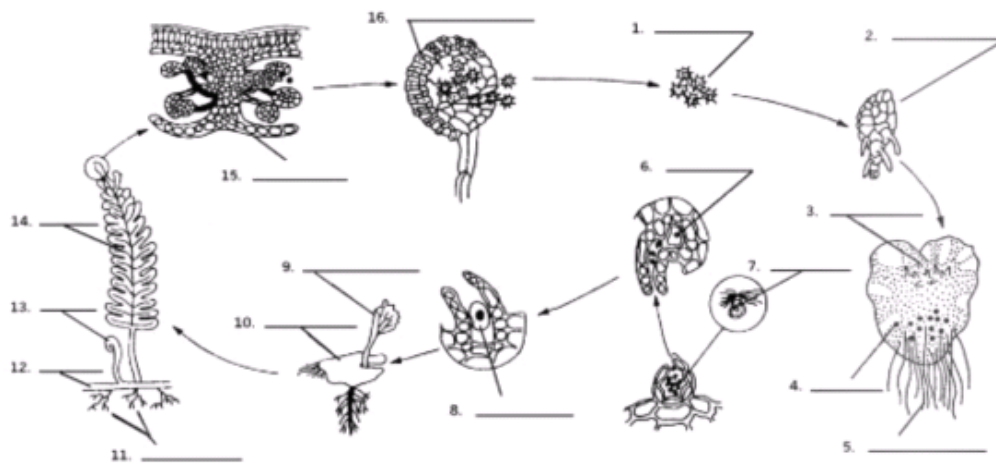


Figure 1:

- **Describe** how the following change occurs: Spores develop within sporophytes that are diploid, but the spores are haploid when released.
- **What** structures hold the gametophytes? Do they partake a significant role in water and mineral absorption?
- **Name** the male and female sex organs. **Explain** why it's important for them to mature at different rates?

Figure 2:

- **Describe** how this cycle differs from Figure 1 in terms of reproduction.
- **Predict** why these plants grow taller than Figure 1.
- **Locate** the sorus by typing the number pointing to them.
- **Describe** where the gametophyte is located during this cycle's "final stage".

Expected Answer:

- **Congratulations, you have finished the botany examination.**
- **Feel free to fill out this test feedback survey: tinyurl.com/RateMySciolyTests.**
- **Interested in Science Olympiad-related classes? Email me, mayurchhиту@gmail.com, for more information and be sure to check out: <https://sciovirtual.org/registration>.**